

Problem Set 2

Solutions by Harsh Bhalala

Problem 1. Failed Induction

(a) By taking $n = 9$, we get $3(9) - 6 = 21$ edges so we have 9 nodes and 21 edges. Now, in our counterexample, two of the 9 nodes have degree 0, and the remaining 7 nodes with 21 edges make a complete K_7 subgraph, which has $\chi(K_7) = 7$, thus proving the claim is false.

(b) Not solved due to lack of Recitation.

Problem 2. High Chromatic Number Without Triangles

(a) for G set of colors $c = \{\text{red, blue, green, black}\}$ and valid coloring function

$$f = \{(w, \text{black}), (v_0, \text{blue}), (v_1, \text{green}), (v_2, \text{blue}), (v_3, \text{green}), (v_4, \text{red}), \\ (u_0, \text{blue}), (u_1, \text{red}), (u_2, \text{blue}), (u_3, \text{green}), (u_4, \text{red})\}$$

Therefore, $\chi(G) \leq 4$.

(b) **Lemma:** For a simple circular graph C with odd number of nodes, $\chi(C) = 3$.

Proof. We know that a graph can only be two colored if it is a bipartite also a bipartite does not contain any odd-numbered cycle, meaning C is not bipartite and cannot be 2-colored, but we can color nodes alternatively with 2 colors and the last node in an odd position with one extra color, thus proving $\chi(C) = 3$. $\square$

For graph G , the subgraph containing the outer nodes $v = u_0, \dots, u_4$ is odd numbered circular graph, and from the lemma we need at least 3 colors; since a 5-cycle can use any single color at most twice, each color must be used in a 2-2-1 distribution. Therefore, any valid coloring of G must match the partial coloring shown.

(c) By using the coloring pattern of part (b) and now let us determine the forced coloring for inner vertices v_0, v_3, v_4 .

- v_0 is adjacent to u_1 (Pink) and u_4 (Green), so it must be colored Blue.
- v_3 is adjacent to u_2 (Blue) and u_4 (Green), so it must be colored Pink.
- v_4 is adjacent to u_0 (Blue) and u_3 (Pink), so it must be colored Green.

We observe that to color all three of them, we at least need 3 colors and now to color the central vertex w we must use 4th color, meaning $\chi(G) > 3$.

Problem 3. Find A Pet

(a)

	Day 1	Day 2	Day 3	Day 4	Day 5	Day 6	Day 7
H	A, B, C, D	D	D	D			
I	$\emptyset$	B, C	C	C			
S	$\emptyset$	A	A, B	B			
Z	$\emptyset$	$\emptyset$	$\emptyset$	A			

(b)

	Day 1	Day 2	Day 3	Day 4	Day 5	Day 6	Day 7
A	I	I					
B	$\emptyset$	Z					
C	S	S					
D	H , Z	H					

(c) Since H prefers D over C and D prefers H the best, they will form a rough matching.

Problem 4. Stable Merriment

(a)

Proof. Let's say for some arbitrary set of stable matchings between n Pokemon and n Trainers on the final day of the Stable Matching Algorithm, there are no merry Trainers. So at most each of the Trainers reject $n - \lceil \frac{n}{2} \rceil - 1$ Pokemon and since there are n Trainers

$$S \leq n \left(n - \lceil \frac{n}{2} \rceil - 1 \right) < \frac{n^2}{2}$$

By elementary algebra

□

(b) Let's say there are no merry Pokemon or Trainers. Since there are no merry Pokemon $S \geq n \lceil \frac{n}{2} \rceil \geq \frac{n^2}{2}$ but that would create a contradiction with part (a) meaning if there are no merry trainers then there must be at least one merry Pokemon and the opposite can be proved similarly

Therefore, we conclude that at least one Pokemon or Trainer must be merry.